



General Chemistry I

Recitation 12

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Dec. 22

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
															Pnictogens	Chalcogens	Halogens			
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026		
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122													5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305													13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

列详细信息

列	第二次期中成绩 ()
满分	10
描述	

统计数据		状态分布		成绩分布	
计数	63	空	2	大于 100	0
最小值	2.60	进行中	0	90 - 100	8
最大值	9.40	需要评分	0	80 - 89	20
范围	6.80	免除	0	70 - 79	17
平均值	7.38			60 - 69	5
中位数	7.70			50 - 59	7
标准偏差	1.62			40 - 49	3
方差	2.62			30 - 39	2
				20 - 29	1
				10 - 19	0
				0 - 9	0
				小于 0	0

Homework6

1. The melting point of TiCl_4 (-24°C) lies below those of TiF_4 (284°C) and TiBr_4 (38°C). Explain why by considering the covalent-ionic nature of these compounds and the intermolecular forces in each case.
2. Show how the glycinate ion ($\text{H}_2\text{N}-\text{CH}_2-\text{COO}^-$) can act as a bidentate ligand. (Draw a Lewis diagram if necessary.) Which atoms in the glycinate ion will bind to a metal ion?
3. Give the chemical formula of each of the following compounds:
 - (a) Silver hexacyanoferrate(II)
 - (b) Potassium tetrakisothiocyanatocobaltate(II)
 - (c) Sodium hexafluorovanadate(III)
 - (d) Potassium trioxalatochromate(III)
4. Draw the structures of all possible isomers for the following complexes. Indicate which isomers are enantiomer pairs.
 - (a) Bromochloro(ethylenediamine)platinum(II) (square-planar)
 - (b) Tetra-amminedichloroiron(III) ion (octahedral)
 - (c) Amminechlorobis(ethylenediamine)iron(III) ion (octahedral)
5. Platinum(IV) forms octahedral complexes. Sketch the structures of all the distinct isomers of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2\text{F}_2]$, indicating which pairs of structures are mirror images of each other

Chemistry of the Transition Metals

Chemistry of the Transition Metals

First (3d) d-block Series (Sc–Zn)

Group	3	4	5	6	7	8	9	10	11	12
Atomic nr	21	22	23	24	25	26	27	28	29	30
Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Electron configuration	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^{10} 4s^1$	$3d^{10} 4s^2$

The energies of the 4s and 3d orbitals are so close: the electron configurations of the neutral atoms and their ions are not easily predicted

Second (4d) d-block Series (Y–Cd)

Atomic nr	39	40	41	42	43	44	45	46	47	48
Element	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Electron configuration	$4d^1 5s^2$	$4d^2 5s^2$	$4d^4 5s^1$	$4d^5 5s^1$	$4d^5 5s^2$	$4d^7 5s^1$	$4d^8 5s^1$	$4d^{10}$	$4d^{10} 5s^1$	$4d^{10} 5s^2$

HSAB Theory (Very important!)

Classification of Lewis Acids and Bases[†]

	Hard	Borderline	Soft
Acids	H^+ , Li^+ , Na^+ , K^+ Be^{2+} , Mg^{2+} , Ca^{2+} Cr^{2+} , Cr^{3+} , Al^{3+} SO_3 , BF_3	Fe^{2+} , Co^{2+} , Ni^{2+} Cu^{2+} , Zn^{2+} , Pb^{2+} SO_2 , BBr_3	Cu^+ , Ag^+ , Au^+ , Tl^+ , Hg^+ Pd^{2+} , Cd^{2+} , Pt^{2+} , Hg^{2+} BH_3
Bases	F^- , OH^- , H_2O , NH_3 CO_3^{2-} , NO_3^- , O^{2-} SO_4^{2-} , PO_4^{3-} , ClO_4^-	NO_2^- , SO_3^{2-} , Br^- N_3^- , N_2 $\text{C}_6\text{H}_5\text{N}$, SCN^-	H^- , R^- , CN^- , CO , I^- CN^- , R_3P , C_6H_6 R_2S

Hard acid (硬酸): High charge density: small, multiply charged ion

Soft acid (软酸): Low charge density: large, low oxidation state

'Hard' applies to species which are small, have high charge states (the charge criterion applies mainly to acids, to a lesser extent to bases), and are weakly polarizable.

'Soft' applies to species which are big, have low charge states and are strongly polarizable

HSAB Theory (Very important!)

Comparing tendencies of hard acids and bases vs. soft acids and bases

Property	Hard acids and bases	Soft acids and bases
atomic/ionic radius	small	large
oxidation state	high	low or zero
polarizability	low	high
electronegativity (bases)	high	low
HOMO energy of bases ^{[9][10]}	low	higher
LUMO energy of acids ^{[9][10]}	high	lower (but more than soft-base HOMO)
affinity	ionic bonding	covalent bonding

“硬亲硬,软亲软,软硬交界都不管”

Solubility: $\text{AgF} \gg \text{AgCl} > \text{AgBr} > \text{AgI} \approx \text{Ag}(\text{CN})_2 > \text{Ag}_2\text{S}$

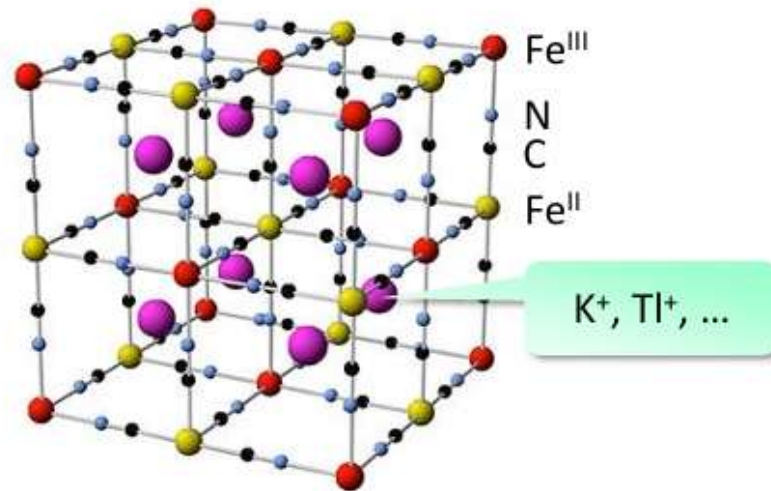
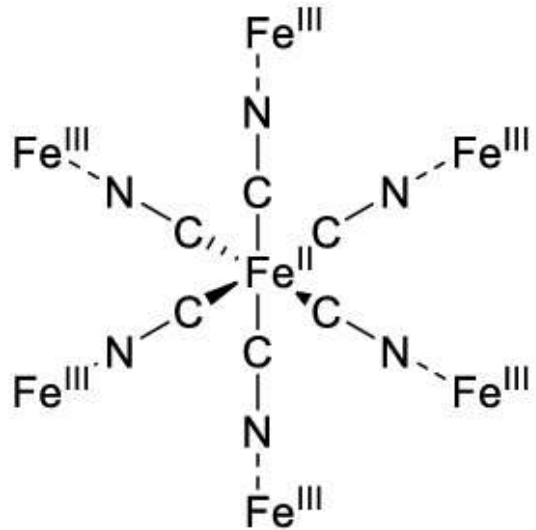
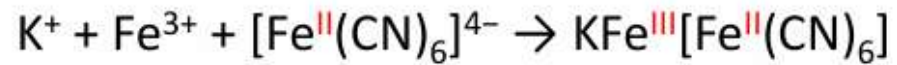
$\text{CsF} > \text{NaF} \gg \text{LiF} > \text{CaF}_2$

Examples of hard and soft acids and bases

Acids				Bases			
hard		soft		hard		soft	
Hydronium	H_3O^+	Mercury	$\text{CH}_3\text{Hg}^+, \text{Hg}_2^{2+}$	Hydroxide	OH^-	Hydride	H^-
Alkali metals	$\text{Li}^+, \text{Na}^+, \text{K}^+$	Platinum	Pt^{2+}	Alkoxide	RO^-	Thiolate	RS^-
Titanium	Ti^{4+}	Palladium	Pd^{2+}	Halogens	F^-, Cl^-	Halogens	I^-
Chromium	$\text{Cr}^{3+}, \text{Cr}^{6+}$	Silver	Ag^+	Ammonia	NH_3	Phosphine	PR_3
Boron trifluoride	BF_3	Borane	BH_3	Carboxylate	CH_3COO^-	Thiocyanate	SCN^-
Carbocation	R_3C^+	P-chloranil	$\text{C}_6\text{Cl}_4\text{O}_2$	Carbonate	CO_3^{2-}	Carbon monoxide	CO
Lanthanides	Ln^{3+}	Bulk metals	M^0	Hydrazine	N_2H_4	Benzene	C_6H_6
Thorium, uranium	$\text{Th}^{4+}, \text{U}^{4+}$	Gold	Au^+				

Soft acids prefer to form bonds with soft bases, whereas hard acids prefer to form bonds with hard bases, all other factors being equal

HSAB Theory (Very important!)

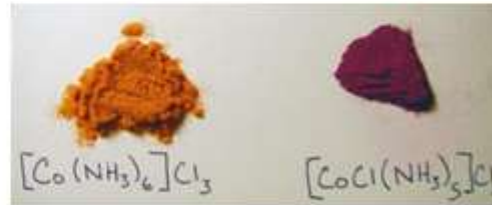


C and N
Which is harder?
What about Fe^{II} and Fe^{III} ?

Introduction to Coordination Chemistry

Compound 1: $\text{CoCl}_3 \cdot 6\text{NH}_3$	Orange-yellow
Compound 2: $\text{CoCl}_3 \cdot 5\text{NH}_3$	Purple
Compound 3: $\text{CoCl}_3 \cdot 4\text{NH}_3$	Green
Compound 4: $\text{CoCl}_3 \cdot 3\text{NH}_3$	Green

Same chemical formulas
Why different properties?



Compound 1: $\text{CoCl}_3 \cdot 6\text{NH}_3$	Orange-yellow
Compound 2: $\text{CoCl}_3 \cdot 5\text{NH}_3$	Purple
Compound 3: $\text{CoCl}_3 \cdot 4\text{NH}_3$	Green
Compound 4: $\text{CoCl}_3 \cdot 3\text{NH}_3$	Green

+ AgNO_3



AgCl

100%
66.7%
33.3%
0%

$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

Formulas may cheat
you sometimes

Werner: **two different kinds of species** associated with the cobalt ions

The primary valences (主价) were anions like Cl^- in simple salts such as CoCl_3 .

The primary valences were nondirectional. (outer coordination sphere 外界)

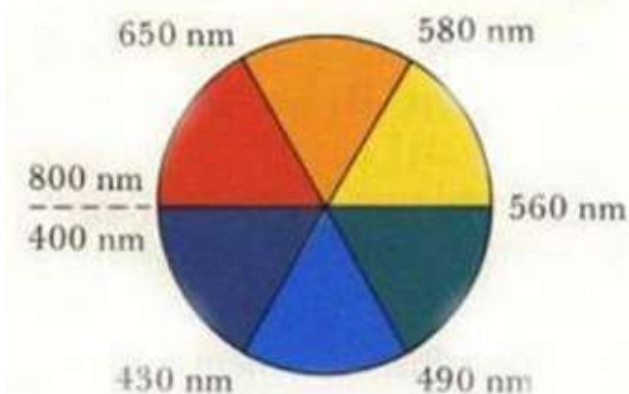
The secondary valences (副价) could be either simple anions or neutral molecules such as NH_3 .

The secondary valences were oriented along well-defined directions. (inner coordination sphere 内界)

Introduction to Coordination Chemistry

Colours of Various Example Coordination Complexes

	Fe ²⁺	Fe ³⁺	Co ²⁺	Cu ²⁺	Al ³⁺	Cr ³⁺
Hydrated Ion	[Fe(H ₂ O) ₆] ²⁺ Pale green Solution	[Fe(H ₂ O) ₆] ³⁺ Yellow/brown Solution	[Co(H ₂ O) ₆] ²⁺ Pink Solution	[Cu(H ₂ O) ₆] ²⁺ Blue Solution	[Al(H ₂ O) ₆] ³⁺ Colourless Solution	[Cr(H ₂ O) ₆] ³⁺ Green Solution
(OH)⁻, dilute	[Fe(H ₂ O) ₄ (OH) ₂] Dark green Precipitate	[Fe(H ₂ O) ₃ (OH) ₃] Brown Precipitate	[Co(H ₂ O) ₄ (OH) ₂] Blue/green Precipitate	[Cu(H ₂ O) ₄ (OH) ₂] Blue Precipitate	[Al(H ₂ O) ₃ (OH) ₃] White Precipitate	[Cr(H ₂ O) ₃ (OH) ₃] Green Precipitate
(OH)⁻, concentrated	[Fe(H ₂ O) ₄ (OH) ₂] Dark green Precipitate	[Fe(H ₂ O) ₃ (OH) ₃] Brown Precipitate	[Co(H ₂ O) ₄ (OH) ₂] Blue/green Precipitate	[Cu(H ₂ O) ₄ (OH) ₂] Blue Precipitate	[Al(OH) ₄] ⁻ Colourless Solution	[Cr(OH) ₆] ³⁻ Green Solution
NH₃, dilute	[Fe(NH ₃) ₆] ²⁺ Dark green Precipitate	[Fe(NH ₃) ₆] ³⁺ Brown Precipitate	[Co(NH ₃) ₆] ²⁺ Straw coloured Solution	[Cu(NH ₃) ₄ (H ₂ O) ₂] ²⁺ Deep blue Solution	[Al(NH ₃) ₃] ³⁺ White Precipitate	[Cr(NH ₃) ₆] ³⁺ Purple Solution
NH₃, concentrated	[Fe(NH ₃) ₆] ²⁺ Dark green Precipitate	[Fe(NH ₃) ₆] ³⁺ Brown Precipitate	[Co(NH ₃) ₆] ²⁺ Straw coloured Solution	[Cu(NH ₃) ₄ (H ₂ O) ₂] ²⁺ Deep blue Solution	[Al(NH ₃) ₃] ³⁺ White Precipitate	[Cr(NH ₃) ₆] ³⁺ Purple Solution
(CO₃)²⁻	FeCO ₃ Dark green Precipitate	Fe ₂ (CO ₃) ₃ Brown Precipitate+bubbles	CoCO ₃ Pink Precipitate	CuCO ₃ Blue/green Precipitate		

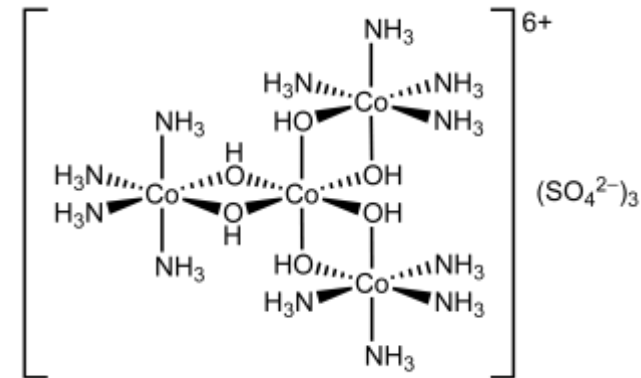
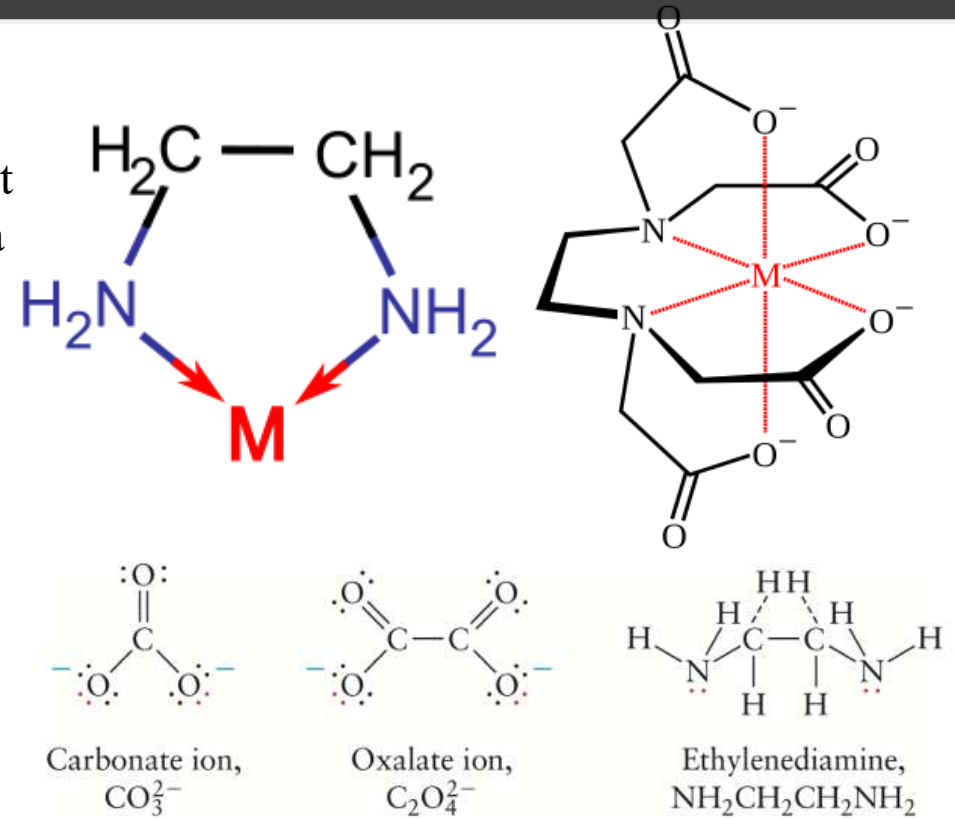
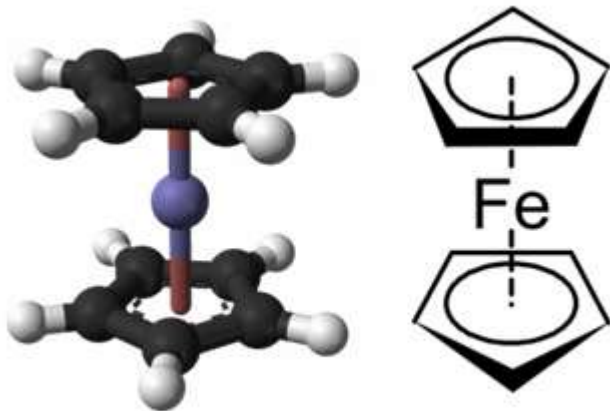


Coordination Chemistry

Coordination can be diverse!

Chelation (螯合) is a type of bonding and sequestration of metal atoms. It involves two or more separate dative covalent bonds between a ligand and a single metal atom, thereby forming a ring structure. The ligand is called a chelant, chelator, chelating agent, or sequestering agent. It is usually an organic compound, but this is not a requirement.

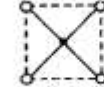
不仅孤对电子可以配位， π 键电子也可以。什么你问sigma键能不能配位？——还真能，但是很少见，表面吸附催化会用到。



Is it chelant?
No.

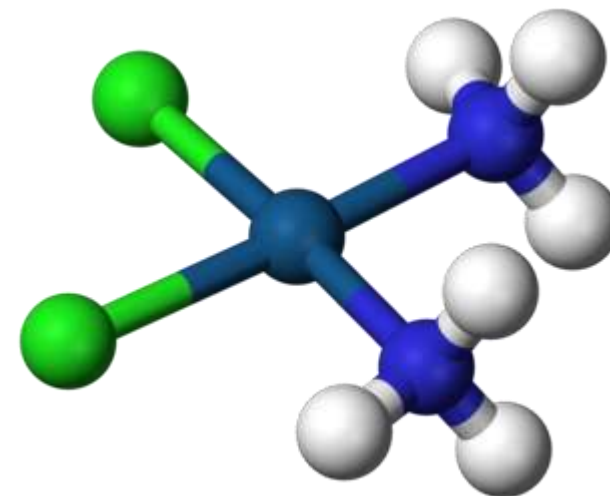
Geometry of Coordination Complex

In coordination chemistry, a structure is first described by its coordination number, the number of ligands attached to the metal (more specifically, the number of donor atoms). Usually one can count the ligands attached, but sometimes even the counting can become ambiguous. Coordination numbers are normally between two and nine, but large numbers of ligands are not uncommon for the lanthanides and actinides. The number of bonds depends on the size, charge, and electron configuration of the metal ion and the ligands. Metal ions may have more than one coordination number.

配位数	2	3	4		5		6	
杂化类型	sp	sp ²	sp ³	dsp ²	dsp ³	d ² sp ²	d ² sp ³	sp ³ d ²
实例	Ag(NH ₃) ₂ ⁺	HgI ₂	FeCl ₄ ⁻	PtCl ₄ ²⁻	Fe(CO) ₅	SbF ₅	Fe(CN) ₆ ³⁻	FeF ₆ ³⁻
立体构型	直线	平面三角	四面体	平面四边形	三角双锥	四角锥	八面体	
模 型								

- Linear for two-coordination
- Trigonal planar for three-coordination
- Tetrahedral or **square planar** for four-coordination
- Trigonal bipyramidal for five-coordination
- Octahedral for six-coordination
- Pentagonal bipyramidal for seven-coordination
- Square antiprismatic for eight-coordination
- Tricapped trigonal prismatic for nine-coordination

Cisplatin, PtCl₂(NH₃)₂, is a coordination complex of platinum(II) with two chloride and two ammonia ligands. It is one of the most successful anticancer drugs.



Sp³d and dsp²

square-planar geometry: is common for four coordinate complexes of Au³⁺, Ir⁺, Rh⁺, and especially common for ions with the d⁸ configurations (Ni²⁺ Pd²⁺ and Pt²⁺)

Attention ! Square planar is allowed in coordination chemistry!

Geometry of Coordination complex

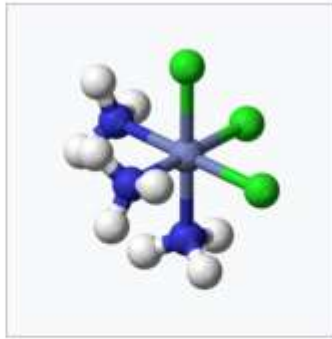
Cis–trans isomerism and facial–meridional isomerism



$cis-[CoCl_2(NH_3)_4]^+$



$trans-[CoCl_2(NH_3)_4]^+$



$fac-[CoCl_3(NH_3)_3]$

面式



$mer-[CoCl_3(NH_3)_3]$

经式

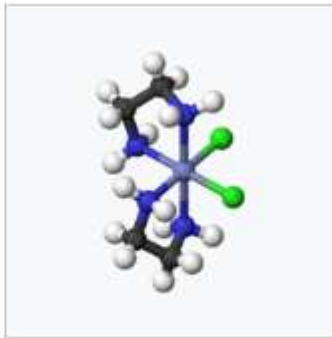
Optical isomerism



$\Lambda-[Fe(ox)_3]^{3-}$



$\Delta-[Fe(ox)_3]^{3-}$



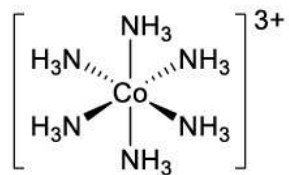
$\Lambda-cis-[CoCl_2(en)_2]^+$



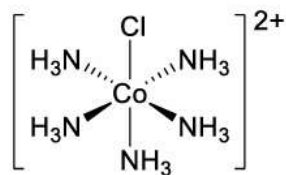
$\Delta-cis-[CoCl_2(en)_2]^+$

Cis–trans isomerism occurs in octahedral and square planar complexes (but not tetrahedral). When two ligands are adjacent they are said to be **cis**, when opposite each other, **trans**. When three identical ligands occupy one face of an octahedron, the isomer is said to be facial, or **fac**. In a *fac* isomer, any two identical ligands are adjacent or *cis* to each other. If these three ligands and the metal ion are in one plane, the isomer is said to be meridional, or **mer**. A *mer* isomer can be considered as a combination of a *trans* and a *cis*, since it contains both trans and cis pairs of identical ligands.

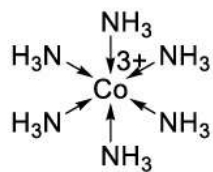
EAN rules (18-electron rule)



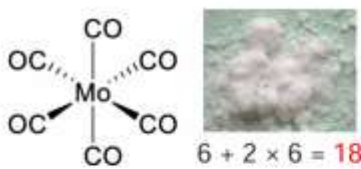
Hexaamminecobalt(III)
六氨合钴(III)



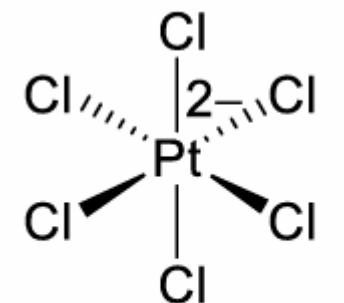
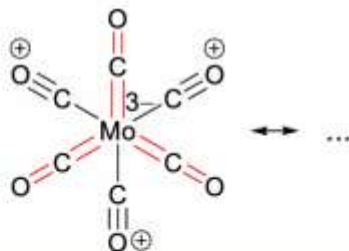
Pentaamminechlorocobalt(III)
一氯五氨合钴(III)



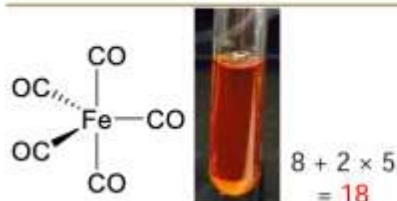
$$9 - 3 + 2 \times 6 = 18$$



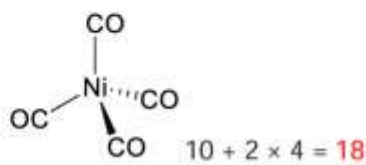
Hexacarbonylmolybdenum(0)
六羰基合钼(0)



$$6 + 2 \times 6 = 18$$



Pentacarbonyliron(0)
五羰基合铁(0)



Tetracarbonylnickel(0)
四羰基合镍(0)

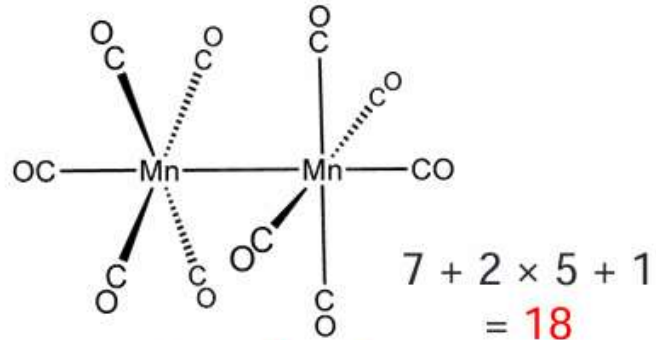
The 18-electron rule is a chemical rule of thumb used primarily for predicting and rationalizing formulas for stable transition metal complexes, especially organometallic compounds. The rule is based on the fact that the valence orbitals in the electron configuration of transition metals consist of five (n-1)d orbitals, one ns orbital, and three np orbitals, where n is the principal quantum number. These orbitals can collectively accommodate 18 electrons as either bonding or non-bonding electron pairs.

$$d + \text{配体} + \text{电荷} = 18!$$

16 17 make sense sometimes.

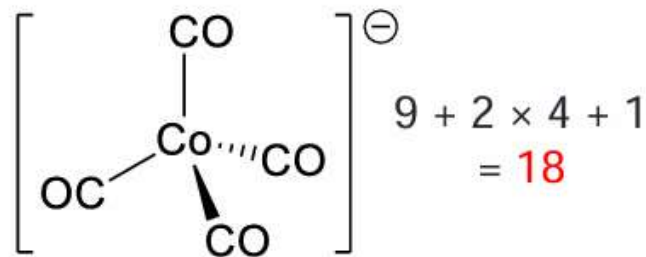
EAN rules (18-electron rule)

Mn: 0



Bis[pentacarbonylmanganese(0)]
双[五羰基合锰(0)]

Co: -1



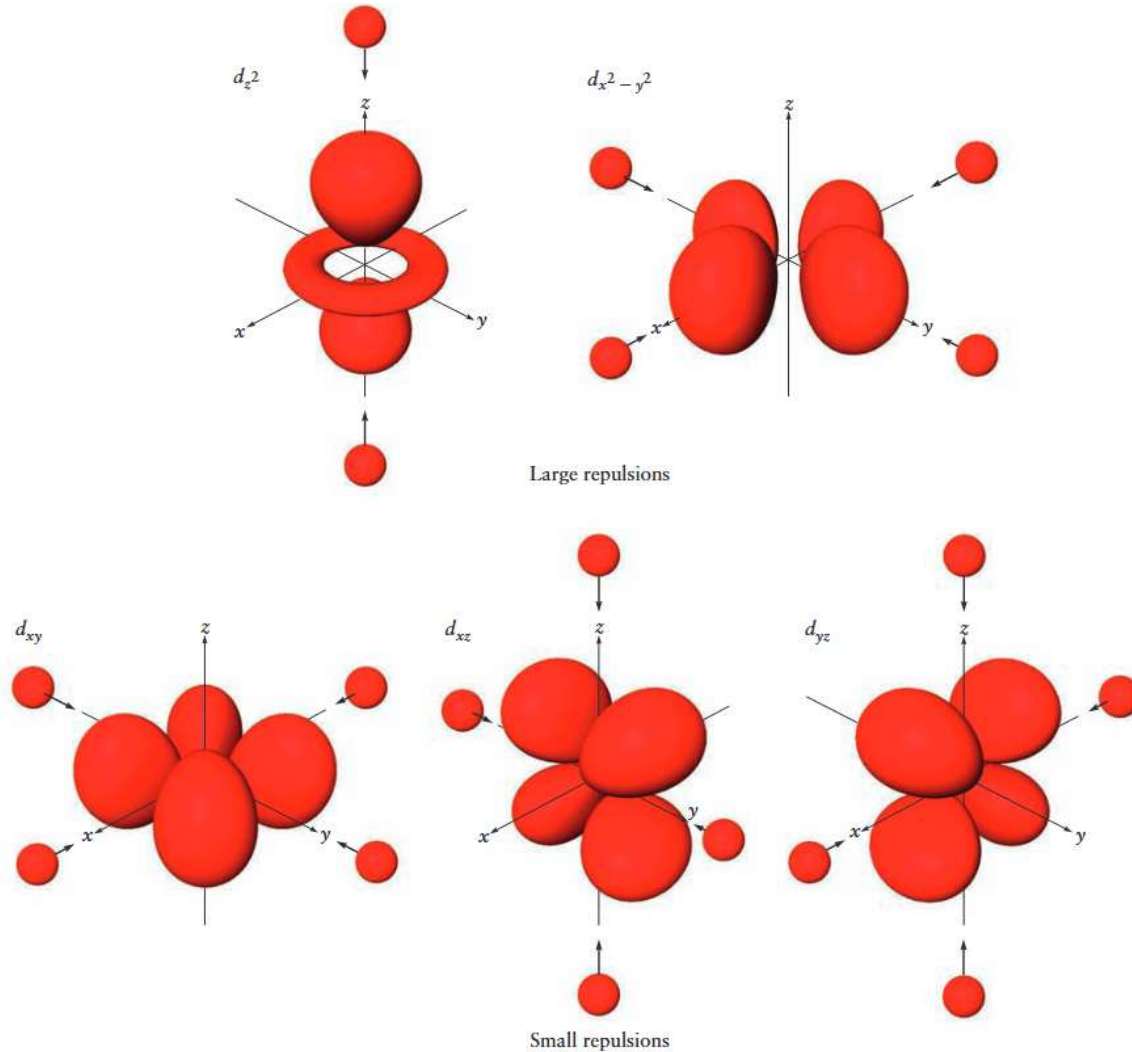
Tetracarbonylcobalt(-1)
四羰基合钴(-1)

The 18-electron rule is a chemical rule of thumb used primarily for predicting and rationalizing formulas for stable transition metal complexes, especially organometallic compounds. The rule is based on the fact that the valence orbitals in the electron configuration of transition metals consist of five (n-1)d orbitals, one ns orbital, and three np orbitals, where n is the principal quantum number. These orbitals can collectively accommodate 18 electrons as either bonding or non-bonding electron pairs.

d+配体+电荷=18!

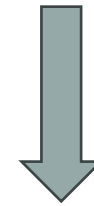
16 17 make sense sometimes.

Crystal Field Theory(Very very very important!!!)



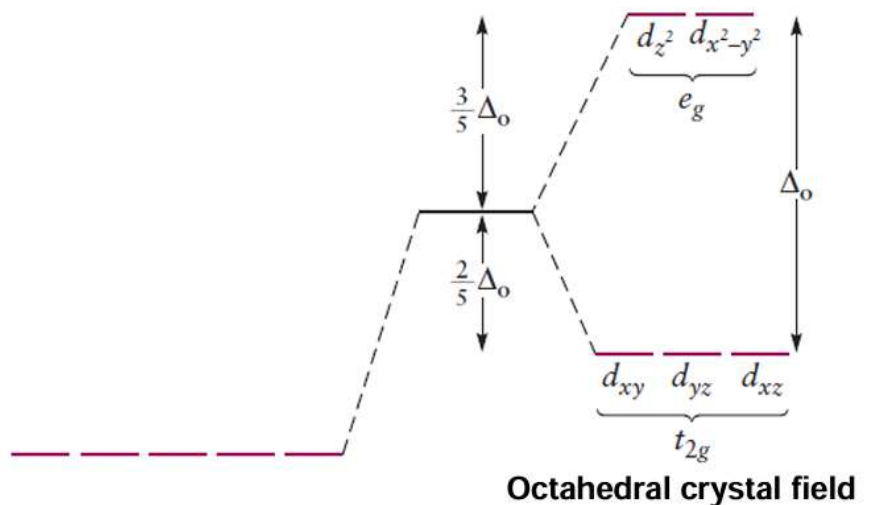
Crystal field theory (CFT) describes the breaking of degeneracies of electron orbital states, usually d or f orbitals, due to a static electric field produced by a surrounding charge distribution (anion neighbors).

- Tetrahedral
- Octahedral



CUBE model will solve that problem

Crystal Field Theory



Δ_o : crystal field splitting energy

In an octahedral field, the energy of the **three t_{2g}** orbitals is lowered by **$2/5 \Delta_o$** , and that of the **two e_g** orbitals is raised by **$3/5 \Delta_o$** relative to the spherical field.

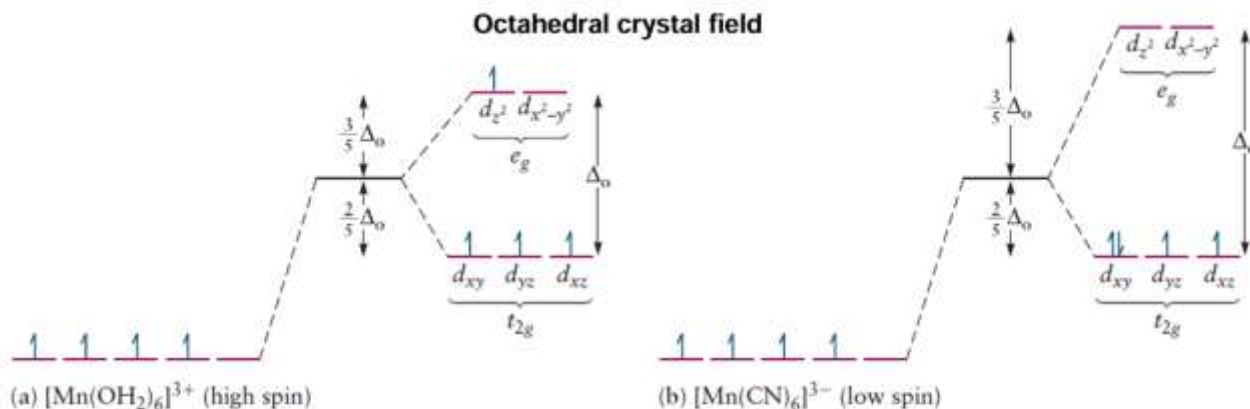


FIGURE 8.18 Electron configuration for (a) high-spin (Δ_o) and (b) low-spin (δ_o) Mn(III) complexes.

Δ_o is small (weak-field ligands):

- high-spin configurations

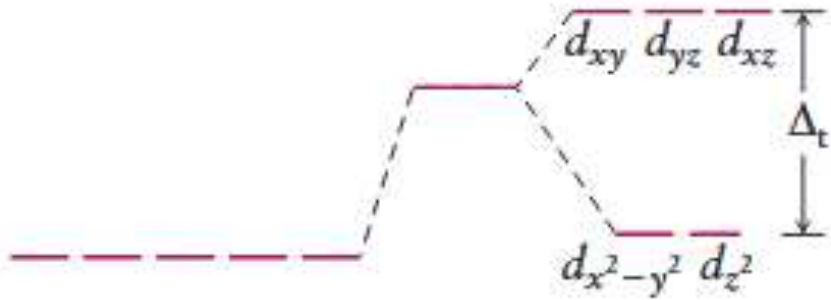
Δ_o is large (strong-field ligands)

- low-spin configurations

Electrons are paired in the lower energy t_{2g} orbital, and the e_g orbitals are not occupied until the t_{2g} levels are filled.

The most common type of complex is **octahedral**, in which six ligands form the vertices of an octahedron around the metal ion. In octahedral symmetry the d -orbitals split into two sets with an energy difference, Δ_{oct} (the **crystal-field splitting parameter**, also commonly denoted by **$10Dq$** for ten times the "differential of quanta") where the d_{xy} , d_{xz} and d_{yz} orbitals will be lower in energy than the d_z^2 and $d_{x^2-y^2}$, which will have higher energy, because the former group is farther from the ligands than the latter and therefore experiences less repulsion. The three lower-energy orbitals are collectively referred to as **t_{2g}** , and the two higher-energy orbitals as **e_g** .

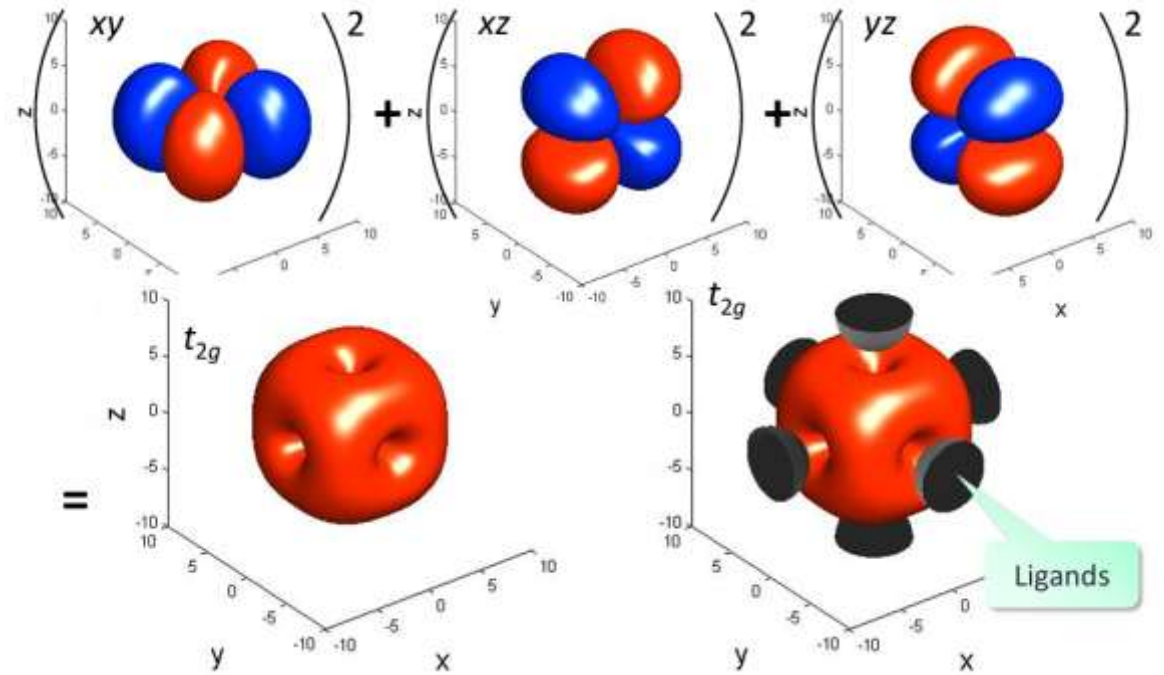
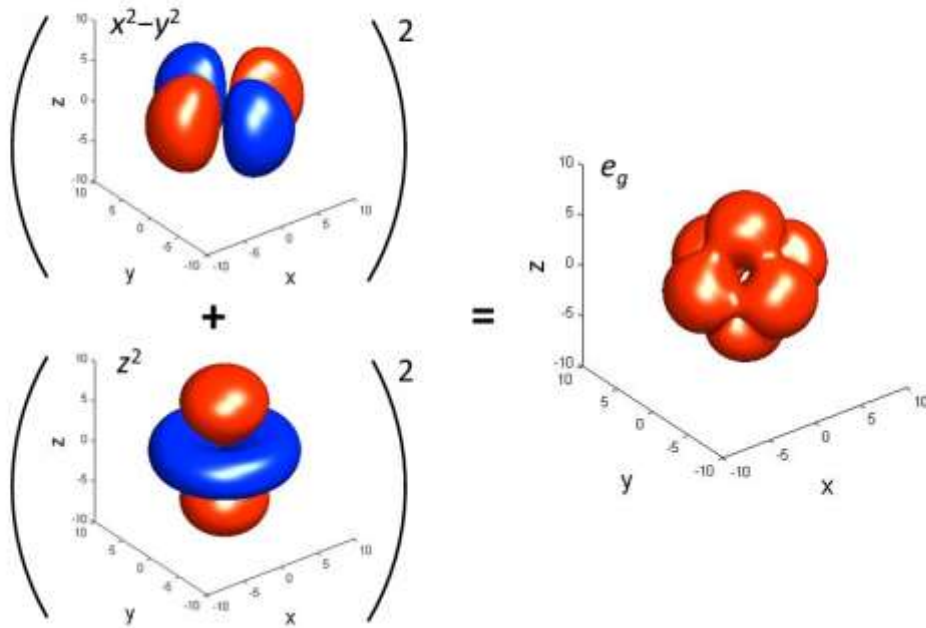
Crystal Field Theory



Tetrahedral complexes are the second most common type; here four ligands form a tetrahedron around the metal ion. In a tetrahedral crystal field splitting, the d -orbitals again split into two groups, with an energy difference of Δ_{tet} . The lower energy orbitals will be d_{z^2} and $d_{x^2-y^2}$, and the higher energy orbitals will be d_{xy} , d_{xz} and d_{yz} - opposite to the octahedral case. Furthermore, since the ligand electrons in tetrahedral symmetry are not oriented directly towards the d -orbitals, the energy splitting will be lower than in the octahedral case. Square planar and other complex geometries can also be described by CFT.

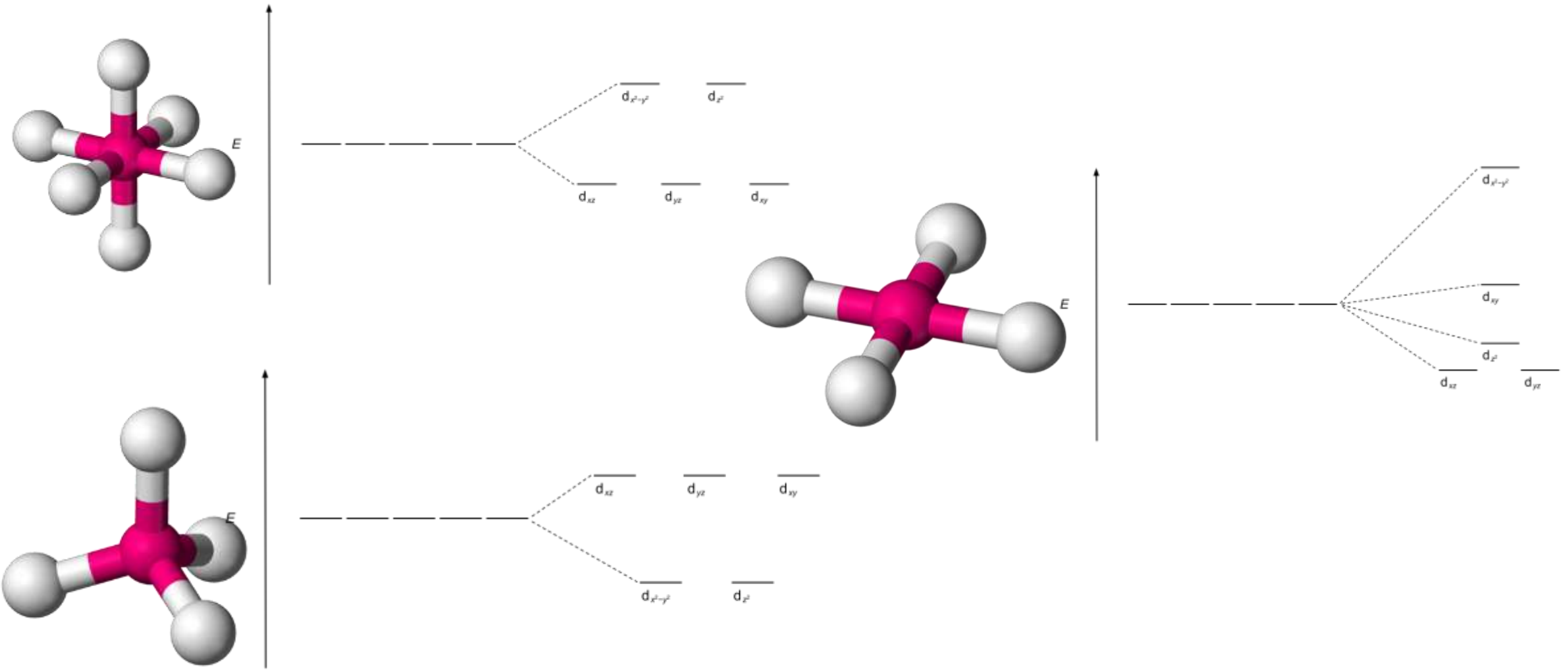
Crystal Field Theory

Which *d* Orbitals?



Bond your partner at first!!!!
Then use the cube model.

Crystal Field Theory



Keypoint:

Repulsion between electrons of central atom and ligands!

Crystal Field Theory

表 4-3 八面体配合物的中心原子 d 电子的组态

	弱 场			强 场	
	d_e			d_e	d_r
d^1	↑			↑	
d^2	↑	↑		↑	↑
d^3	↑	↑	↑	↑	↑
d^4	↑	↑	↑	↑↓	↑
d^5	↑	↑	↑	↑↓	↑↓
d^6	↑↓	↑	↑	↑↓	↑↓
d^7	↑↓	↑↓	↑	↑↓	↑↓
d^8	↑↓	↑↓	↑↓	↑↓	↑↓
d^9	↑↓	↑↓	↑↓	↑↓	↑↓
d^{10}	↑↓	↑↓	↑↓	↑↓	↑↓

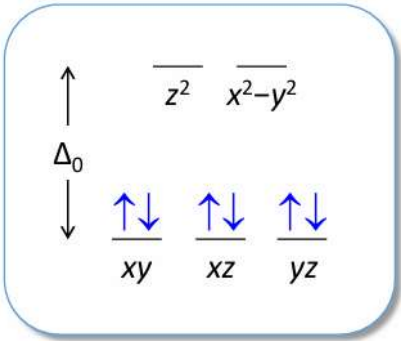
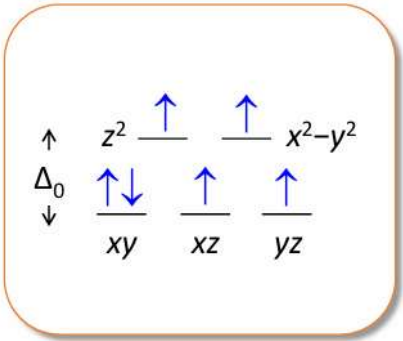
先确定裂分状态
填入中心原子的d电子（依据强弱场规则）
注意d 4-7 情况
计算能量

强场配体会倾向于克服电子成对能，将电子“挤”成对

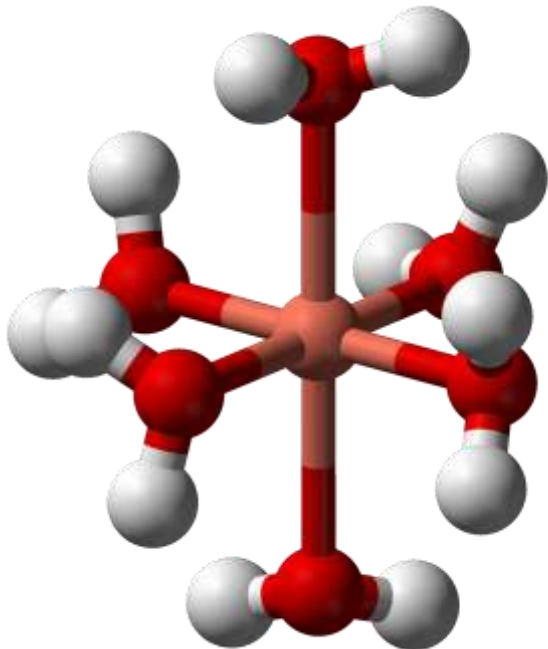
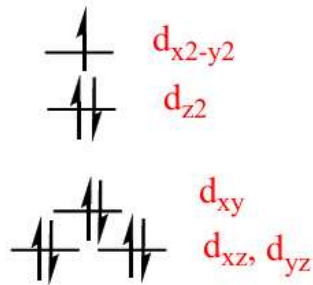
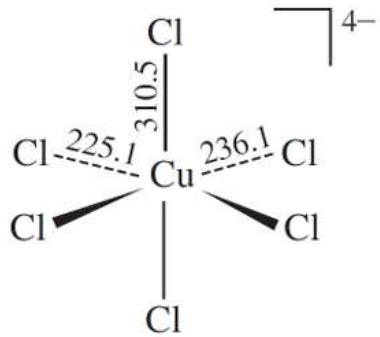
$I^- < Br^- < S_2^- < SCN^- < Cl^- < NO_3^- < N_3^- < F^- < OH^- < OH_2 < NCS^- < py < NH_3 < en < NO_2^- < PPh_3 < CN^- \approx CO \approx NO^+$
Weak-field ligands (high-spin) Intermediate-field ligands Strong-field ligands (low-spin)

Configuration		d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
Examples		Ti^{3+}	Ti^{2+}, V^{3+}	V^{2+}, Cr^{3+}	Cr^{2+}, Mn^{3+}	Mn^{2+}, Fe^{3+}	Fe^{2+}, Co^{3+}	Co^{2+}, Ni^{3+}	Ni^{2+}, Pt^{2+}	Cu^{2+}	Zn^{2+}
HIGH SPIN	e_g	—	—	—	↑	↑↑	↑↑	↑↑	↑↑	↑↓↑	↑↓↑↓
	t_{2g}	↑	↑↑	↑↑↑	↑↑↑	↑↑↑	↑↓↑↑	↑↓↑↓	↑↓↑↓	↑↓↑↓	↑↓↑↓
	CFSE	$-\frac{2}{5} \Delta_o$	$-\frac{4}{5} \Delta_o$	$-\frac{6}{5} \Delta_o$	$-\frac{3}{5} \Delta_o$	0	$-\frac{2}{5} \Delta_o$	$-\frac{4}{5} \Delta_o$	$-\frac{6}{5} \Delta_o$	$-\frac{3}{5} \Delta_o$	0
LOW SPIN	e_g				—	—	—	↑			
	t_{2g}				↑↑↑	↑↓↑↓	↑↓↑↓	↑↓↑↓			
	CFSE	Same as high spin			$-\frac{8}{5} \Delta_o$	$-\frac{10}{5} \Delta_o$	$-\frac{12}{5} \Delta_o$	$-\frac{9}{5} \Delta_o$	Same as high spin		

CFSE, Crystal field stabilization energies.



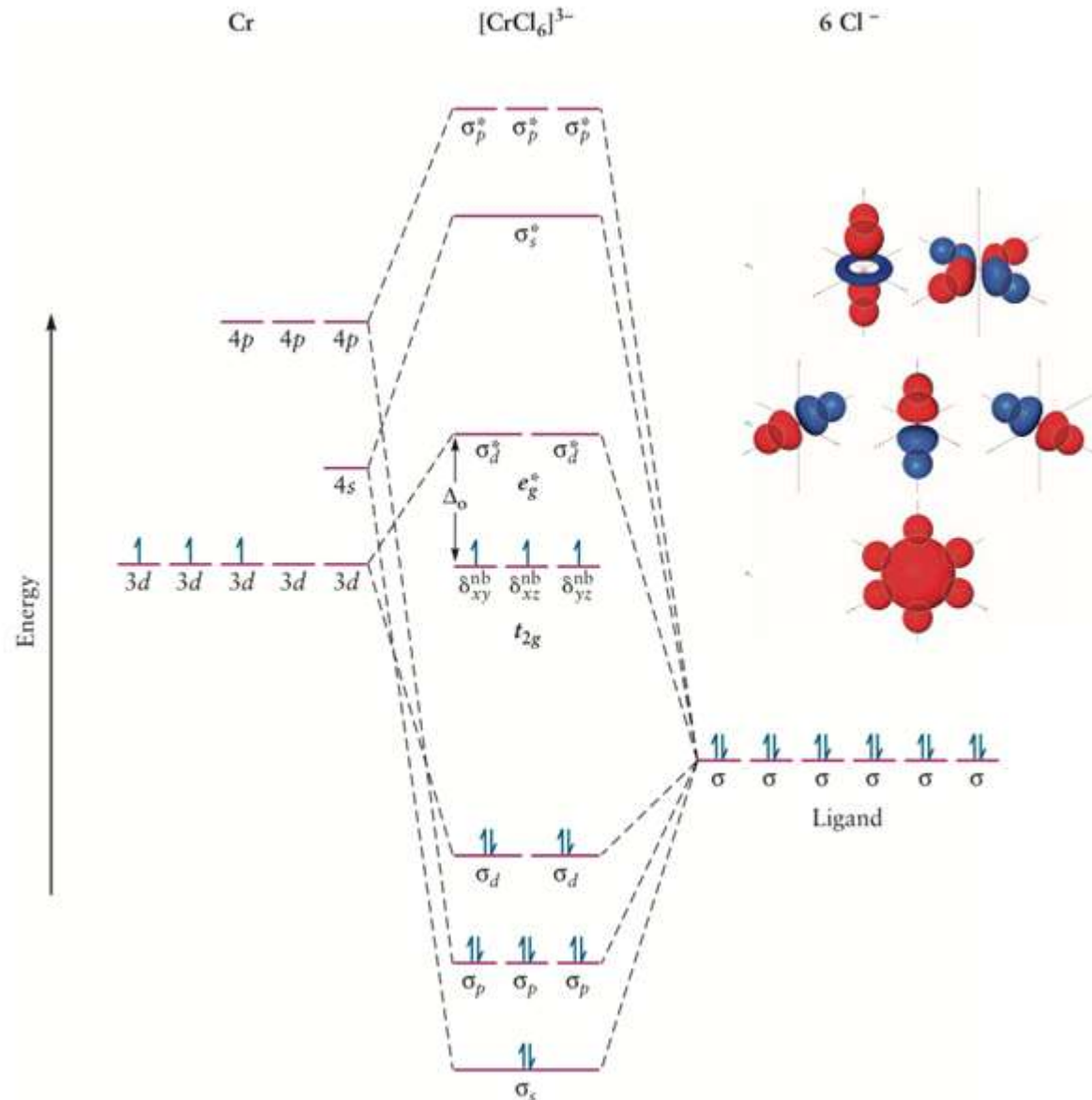
Jahn-Teller Effect



The **Jahn-Teller effect**, sometimes also referred to as **Jahn-Teller distortion**, describes the geometrical distortion of **molecules** and **ions** that results from certain electron configurations. The **Jahn-Teller theorem** essentially states that any non-linear molecule with a spatially **degenerate** electronic ground state will undergo a geometrical distortion that removes that degeneracy, because the distortion lowers the overall energy of the species.

In octahedral complexes, the Jahn-Teller effect is most pronounced when an **odd number** of electrons occupy the e_g orbitals. This situation arises in complexes with the configurations **d^9 , low-spin d^7 or high-spin d^4** complexes, all of which have doubly degenerate ground states.

MO and Crystal Field Theory



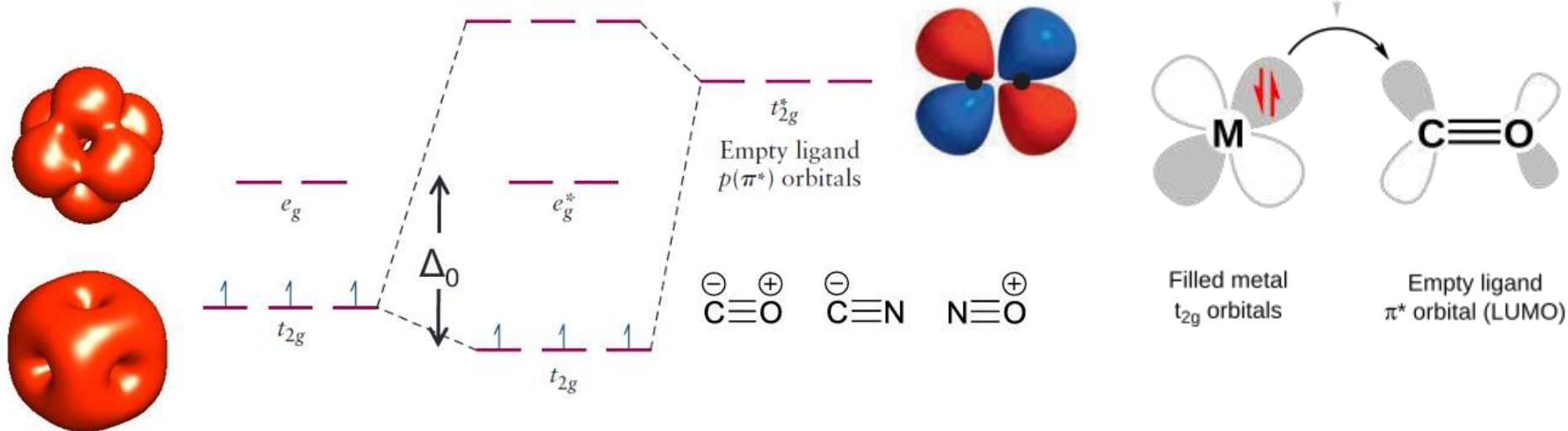
Ligand-to-metal ($L \rightarrow M$) σ donation

Strong interactions decrease the energy of (**stabilize**) the **bonding σ_d** orbital but increase the energy of (**de-stabilize**) the **σ_d^* orbital**.

This description explains the increase in Δ_o as the strength of the metal–ligand interaction increases.

MO and Crystal Field Theory

Strongest Ligands: π Acceptors

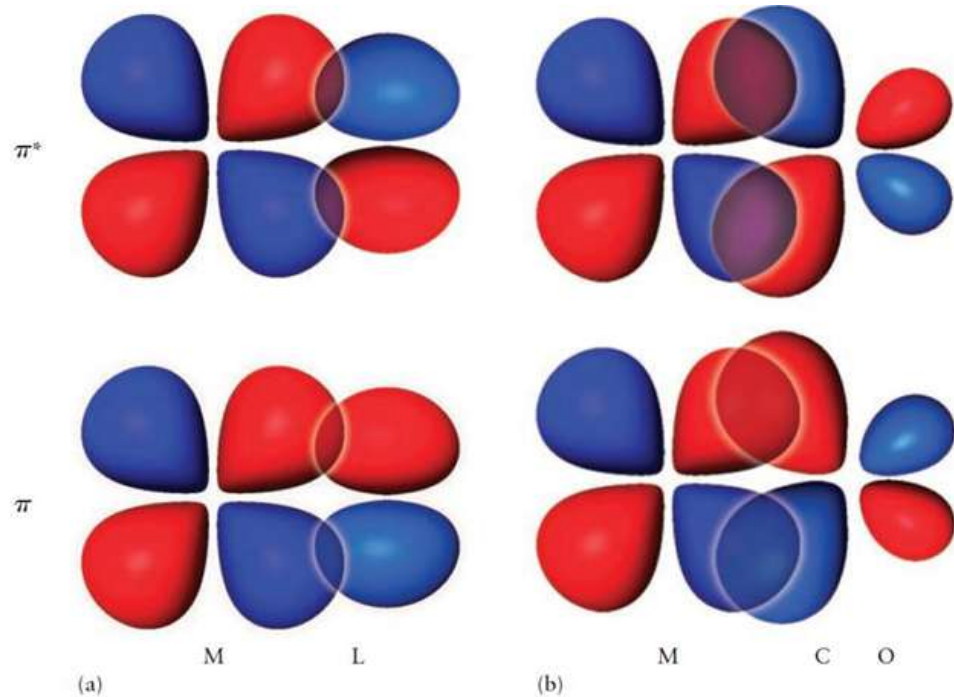


π bonding in octahedral complexes occurs in two ways: via any ligand p -orbitals that are not being used in σ bonding, and via any π or π^* molecular orbitals present on the ligand.

One important π bonding in coordination complexes is **metal-to-ligand π bonding**, also called **π backbonding**. It occurs when the **LUMOs** (lowest unoccupied molecular orbitals) of the ligand are anti-bonding π^* orbitals. These orbitals are close in energy to the d_{xy} , d_{xz} and d_{yz} orbitals, with which they combine to form bonding orbitals (i.e. orbitals of lower energy than the aforementioned set of d -orbitals). The corresponding anti-bonding orbitals are higher in energy than the anti-bonding orbitals from σ bonding so, after the new π bonding orbitals are filled with electrons from the metal d -orbitals, Δ_o has increased and the bond between the ligand and the metal strengthens. The ligands end up with electrons in their π^* molecular orbital, so the corresponding **π bond within the ligand weakens**.

MO and Crystal Field Theory

Strongest Ligands: π Acceptors



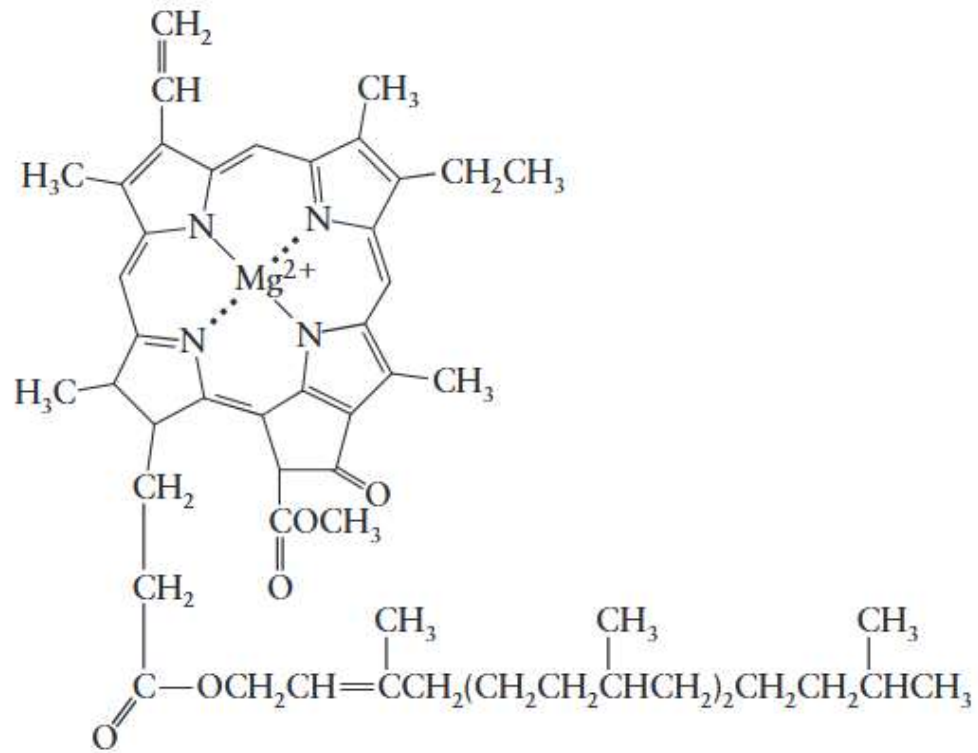
CO and CN^- are strongest ligands

$\text{I}^- < \text{Br}^- < \text{S}_2^- < \text{SCN}^- < \text{Cl}^- < \text{NO}_3^- < \text{N}_3^- < \text{F}^- < \text{OH}^- < \text{OH}_2 < \text{NCS}^- < \text{py} < \text{NH}_3 < \text{en} < \text{NO}_2^- < \text{PPh}_3 < \text{CN}^- \approx \text{CO} \approx \text{NO}^+$

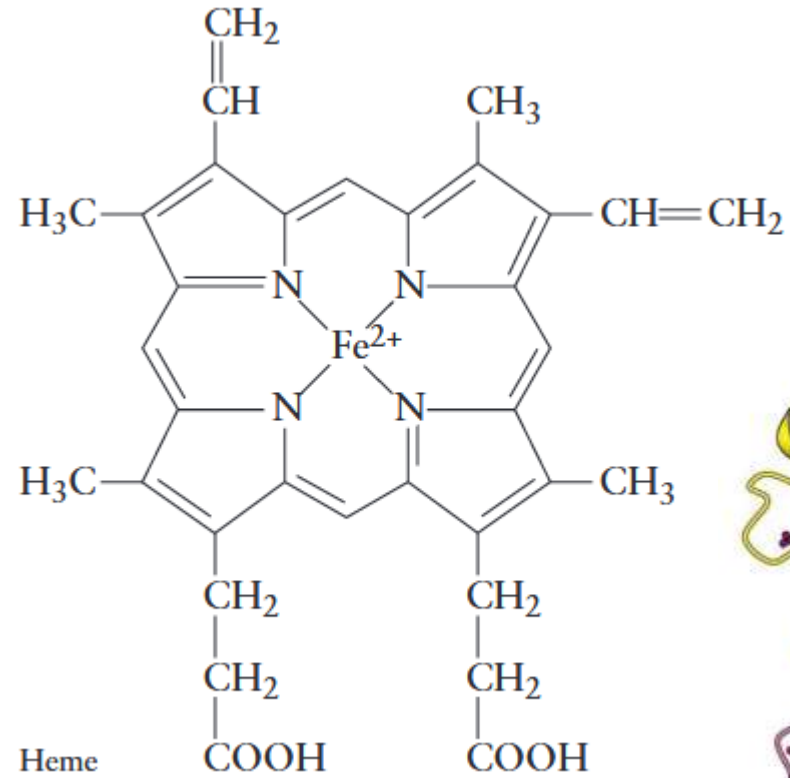
Weak-field ligands (high-spin) Intermediate-field ligands Strong-field ligands (low-spin)

The greater stabilization that results from metal-to-ligand bonding is caused by the donation of negative charge away from the metal ion, towards the ligands. This allows the metal to accept the σ bonds more easily. The combination of ligand-to-metal σ -bonding and metal-to-ligand π -bonding is a synergic effect, as each enhances the other.

Coordination and Biology



Chlorophyll a



Heme

Hemoglobin

CO?

